

3W USB Stereo Audio Amplifier

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A vast range of audio-associated integrated circuits (ICs) are readily available for use by experimenters, professional design engineers and hobbyists. Most of the ICs are modestly priced with universally recognised part numbers. Described here is an interesting circuit of a 3W USB stereo amplifier based on PAM8403 IC from Power Analog Microelectronics. Designed for portable applications, PAM8403 is a 3W (class-D) stereo amplifier suitable for high-quality sound reproduction. Its filter-less architecture allows the chip to drive inductive loads directly, without any low-pass output filters, which saves the system cost and size.

Circuit and working

Fig. 1 shows circuit diagram of the 3W USB stereo audio amplifier. It is built around PAM8403 module (Board1), 5V voltage regulator LM1117-5 (IC1) and a

few other components.

PAM8403 IC is available in DIP-16 (regular) and SOP-16 (SMD) packages. With a little skill and patience, even a novice can rig up the stereo amplifier system on a small prototyping board. DIP-16 package may not be available from a local component vendor as the package is not very popular in this age of microelectronics. Fortunately, PAM8403 based stereo audio amplifier modules in pre-wired form are readily available from online component vendors at affordable price. Anyone can build one's own stereo audio system using this module just by adding some external components.

The circuit shown here may be assembled on a separate prototyping board. Use of suitable pin headers is recommended for mounting the PAM8403 module on prototyping board. It is seen from the circuit diagram that two separate power inputs, USB B-type (CON5) and external 7-12V DC (Jack2), have been incorporated for

total convenience and flexibility. However, if you are interested only in USB power, the onboard regulator (and related components) is not needed. Anyway, before giving power to the circuit, the entire assembly must be checked carefully for correct connections.

Note that the absolute maximum working voltage of PAM8403 module is 5.5V. Therefore do not use four AA dry cells ($1.5V \times 4 = 6V$) to power the system. It is recommended to use four rechargeable (Ni-Cd/Ni-Mh) cells ($1.2V \times 4 = 4.8V$), or three dry/alkaline cells ($1.5V \times 3 = 4.5V$). Further, do not make the audio input signals too high as that will deteriorate the sound quality, and probably destroy the chip

Test Points

Test point	Details
TP0	0V
TP1	5V when USB is used
TP2	5V when Jack2 is used

PARTS LIST

Semiconductors:

- Board1 - PAM8403 module
- IC1 - 5V regulator LM1117-5
- D1, D2 - 1N5819 diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- VR1, VR2 - 50k potentiometer

Capacitors:

- C1, C2 - 10 μ F, 25V electrolytic

Miscellaneous:

- S1 - On/off switch
- Jack1 - 3.5mm stereo input jack
- Jack2 - DC power jack
- CON1 - 3-pin connector
- CON2-CON4 - 2-pin connector
- CON5 - USB B-type connector
- LS1, LS2 - 4-ohm, 3W speaker

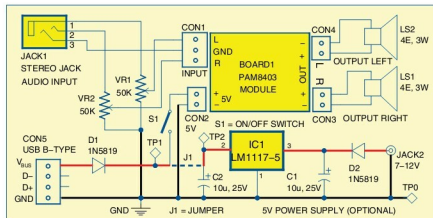


Fig. 1: Circuit diagram of 3W USB stereo audio amplifier

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